

INDUSTRIAL AUTOMATION ENGINEERING PORTFOLIO

Project 6 — Version 2

Industrial Conveyor Sorting System

PLC / Simulation Portfolio Project

Version 2 — Structured Controls Upgrade

1. Project Overview

Project 6 Version 2 develops the original conveyor project into a PLC-controlled industrial conveyor sorting system with automatic reject handling. I designed the system to transport products, track them from an entry station to a sorting station, retain their inspection classification, automatically reject defective products, verify successful rejection, detect abnormal process conditions, and maintain production counts.

The final program was developed for an Allen-Bradley MicroLogix 1100 PLC using RSLogix Micro Starter Lite. It separates machine permission, operator run request, automatic and manual conveyor commands, product tracking, reject sequencing, process diagnostics, fault recovery, and production counting rather than controlling the conveyor and reject actuator directly from field inputs.

The final ladder contains Rungs 0000–0028 followed by END at Rung 0029. The final screenshot is the technical source of truth for the addresses, branches, timer settings, instructions, and implemented behavior documented in this portfolio.

This project was developed and tested in a PLC training/simulation environment. The testing demonstrates PLC program behavior and should not be interpreted as physical conveyor commissioning, pneumatic-system validation, or machine-safety certification.

2. Engineering Objective

- Prevent operation when Auto and Manual modes are selected simultaneously.
- Monitor E-stop, conveyor overload, and air-pressure healthy conditions.
- Separate primary machine faults from process-generated faults.
- Latch abnormal conditions for deliberate operator recovery.
- Clear abandoned product/reject states during controlled reset.
- Establish centralized Start_Permissive and System_Ready states.
- Separate automatic and manual conveyor requests from the physical motor output.
- Apply a controlled conveyor startup delay.
- Track one product between entry and sorting stations.
- Store the reject classification independently from the inspection input.
- Detect excessive product travel time.
- Stop automatic conveyor movement during the reject sequence.
- Operate the reject actuator through a timed sequence.
- Verify successful rejection with Reject_Confirm.
- Generate a reject timeout fault when confirmation is missing.
- Count total processed products and maintain separate accepted and rejected totals.

3. Functional Requirements

- Detect simultaneous Auto_Mode I:1/3 and Manual_Mode I:1/4 through Mode_Conflict B3:0/3.
- Latch Fault_Active B3:0/4 from Mode Conflict, unhealthy EStop_OK I:1/5, unhealthy Overload_OK I:1/6, unhealthy Air_Pressure_OK I:1/11, Jam_Fault B3:1/1, or Reject_Timeout_Fault B3:1/2.
- Allow fault reset only through the implemented Reset_PB I:1/2 healthy-condition chain.
- Clear Reject_Sequence B3:1/0, Product_Tracked B3:0/6, and Reject_Product B3:0/7 through the separate sequence recovery rung.
- Generate Start_Permissive B3:0/1 when the implemented operating conditions are healthy.
- Latch Machine_Run_Request B3:0/0 from Start_PB I:1/0 and Start_Permissive.
- Unlatch Machine_Run_Request from Stop_PB I:1/1, Fault_Active, or loss of Start_Permissive.

- Generate System_Ready B3:0/2 when Machine_Run_Request and Start_Permissive are true and either Auto or Manual mode is selected.
- Generate Conveyor_Request B3:0/5 in automatic operation when System_Ready and Auto_Mode are true and Reject_Sequence is false.
- Generate Manual_Conveyor_Request B3:1/3 from System_Ready, Manual_Mode, and Manual_Conveyor_PB I:1/12.
- Use T4:0 with time base 1.0 and preset 3 as the conveyor startup delay.
- Energize Conveyor_Motor O:2/0 only after T4:0/DN and while an automatic or manual conveyor request remains present.
- Latch Product_Tracked B3:0/6 when Entry_Photoeye I:1/7 detects a product during valid automatic operation.
- Latch Reject_Product B3:0/7 when Inspection_Reject I:1/8 classifies the tracked product for rejection.
- Use Jam_Timer T4:1 with time base 1.0 and preset 120 to monitor travel to Sort_Photoeye I:1/9.
- Latch Jam_Fault B3:1/1 when the jam timer is done and the tracked product still has not reached the sort station while the conveyor is running.
- Generate Accepted_Event B3:1/5 for a tracked non-reject product at Sort_Photoeye.
- Latch Reject_Sequence B3:1/0 for a tracked reject product at Sort_Photoeye.
- Use Reject_Extend_Timer T4:2 with time base 0.01 and preset 1000.
- Generate Manual_Reject_Request B3:1/4 from System_Ready, Manual_Mode, Manual_Reject_PB I:1/13, and XIO Fault_Active.
- Energize Reject_Solenoid O:2/1 from the automatic reject sequence or manual reject request while Fault_Active is false.
- Generate Reject_Complete_Event B3:1/6 when Reject_Sequence, T4:2/DN, and Reject_Confirm I:1/10 are true.
- Latch Reject_Timeout_Fault B3:1/2 when Reject_Confirm is absent after the reject timer is done.
- Count total products with C5:0 from Accepted_Event or Reject_Complete_Event.
- Increment accepted total N7:0 from Accepted_Event and rejected total N7:1 from Reject_Complete_Event.
- Perform accepted/rejected cleanup after the counting rungs.

4. System Components

Component	Address	Function
Start Push Button	I:1/0	Machine start command
Stop Push Button	I:1/1	Machine stop command
Reset Push Button	I:1/2	Fault and sequence recovery
Auto Mode	I:1/3	Automatic production mode
Manual Mode	I:1/4	Manual maintenance mode
EStop_OK	I:1/5	E-stop healthy feedback
Overload_OK	I:1/6	Conveyor overload healthy feedback
Entry Photoeye	I:1/7	Product entry detection
Inspection Reject	I:1/8	Reject classification input
Sort Photoeye	I:1/9	Sorting-station product detection
Reject Confirm	I:1/10	Reject confirmation feedback
Air Pressure OK	I:1/11	Pneumatic utility healthy feedback
Manual Conveyor PB	I:1/12	Hold-to-run manual conveyor
Manual Reject PB	I:1/13	Hold-to-run manual reject
Conveyor Motor	O:2/0	Conveyor motor command
Reject Solenoid	O:2/1	Pneumatic reject command
Startup Delay	T4:0	3-second conveyor startup delay
Jam Timer	T4:1	120-second simulation travel monitor
Reject Extend Timer	T4:2	10-second programmed reject interval
Total Product Count	C5:0	Total processed-product counter

Accepted Count	N7:0	Accepted-product integer total
Rejected Count	N7:1	Rejected-product integer total

5. PLC Data and I/O Assignment

Address	Symbolic Name	Type / Purpose
B3:0/0	Machine_Run_Request	Latched machine run request
B3:0/1	Start_Permissive	Machine operating permission
B3:0/2	System_Ready	Ready/running state
B3:0/3	Mode_Conflict	Auto/Manual conflict diagnostic
B3:0/4	Fault_Active	Latched master fault
B3:0/5	Conveyor_Request	Automatic conveyor request
B3:0/6	Product_Tracked	Stored active-product state
B3:0/7	Reject_Product	Stored reject classification
B3:1/0	Reject_Sequence	Automatic reject sequence state
B3:1/1	Jam_Fault	Latched product-travel fault
B3:1/2	Reject_Timeout_Fault	Latched missing-confirmation fault
B3:1/3	Manual_Conveyor_Request	Manual conveyor request
B3:1/4	Manual_Reject_Request	Manual reject request
B3:1/5	Accepted_Event	Accepted-product completion event
B3:1/6	Reject_Complete_Event	Confirmed reject completion event
T4:0	Delay_Timer	TON, TB 1.0, PRE 3
T4:1	Jam_Timer	TON, TB 1.0, PRE 120
T4:2	Reject_Extend_Timer	TON, TB 0.01, PRE 1000
C5:0	Total_Product_Count	CTU
N7:0	Accepted Product Total	Integer
N7:1	Rejected Product Total	Integer

6. Control Strategy

I organized the program around the control path Machine Conditions → Start Permission → Run Request → System Ready → Conveyor Control → Product Tracking → Classification → Accepted/Reject Processing → Counting → Cleanup.

The first part of the ladder establishes whether the machine is allowed to operate. Mode conflict, E-stop feedback, overload feedback, air-pressure feedback, jam faults, and reject timeout faults feed the master fault architecture. Start_Permissive B3:0/1 consolidates the principal operating conditions, while Machine_Run_Request B3:0/0 stores the operator Start command.

Automatic and manual conveyor requests are kept separate from the physical Conveyor_Motor O:2/0 output. Either request can start T4:0, and the motor is energized only after the programmed delay. In automatic sorting, Product_Tracked B3:0/6 remembers the product between stations and Reject_Product B3:0/7 carries its inspection result.

At the sorting station, an acceptable product generates Accepted_Event B3:1/5. A reject product instead latches Reject_Sequence B3:1/0, stops automatic conveyor movement, operates the timed reject sequence, and requires Reject_Confirm I:1/10 before Reject_Complete_Event B3:1/6 is generated. Counting occurs before cleanup so the PLC scan order preserves production totals.

7. Engineering Design Decisions

Design Decision	Engineering Purpose
Dedicated Mode Conflict	Detect contradictory operating-mode selection
Latched Fault Active	Preserve abnormal conditions for acknowledgement
Separate primary/process fault rungs	Improve diagnostic organization
Conditional fault reset	Prevent reset while required healthy conditions are absent
Separate sequence recovery	Clear abandoned product/reject states
Start Permissive	Centralize machine availability
Latched Machine Run Request	Store momentary Start command
System Ready separated from run request	Improve diagnostics
Separate Auto/Manual conveyor requests	Keep operating modes organized
Startup timer	Provide controlled conveyor startup
Product Tracked latch	Remember product between sensors
Reject Product latch	Carry inspection result downstream
Jam timer and recheck	Detect abnormal product travel
Reject Sequence latch	Maintain reject operation after sort sensor changes
Reject confirmation	Verify actual process completion
Reject timeout fault	Detect missing confirmation
Single Reject Solenoid output rung	Avoid duplicate output coils
Event bits before counting/cleanup	Separate completion, counting, and state clearing
Separate accepted/rejected totals	Provide production statistics

8. Ladder Logic Description

8.1 Rung 0000 — Mode Conflict Detection

Ladder Logic

```
|----[ XIC I:1/3 ]----[ XIC I:1/4 ]----- ( OTE B3:0/3 )----|
      Auto_Mode      Manual_Mode                Mode_Conflict
```

Rung Description:

Rung 0000 monitors the Auto and Manual mode selectors. If Auto_Mode I:1/3 and Manual_Mode I:1/4 are ON at the same time, Mode_Conflict B3:0/3 becomes true. Because the rung uses an OTE instruction, the bit follows the current selector condition and turns back off when the conflict is removed.

Engineering Purpose:

I included this check because the conveyor should not accept Automatic and Manual operation at the same time. Mode_Conflict is used by the fault and permissive logic so an invalid mode selection prevents normal operation.

8.2 Rung 0001 — Primary Fault Detection

Ladder Logic

```
+----[ XIC B3:0/3 ]-----+
|      Mode_Conflict      |
+----[ XIO I:1/5 ]-----+
|      EStop_OK           |
|----+----[ XIO I:1/6 ]-----+---- ( OTL B3:0/4 )----|
      Overload_OK                Fault_Active
```

Rung 0001 combines three primary abnormal conditions into the master fault. Mode_Conflict B3:0/3 makes the first branch true, while the XIO instructions detect loss of the healthy EStop_OK I:1/5 or Overload_OK I:1/6 signals. If any branch becomes true, the OTL instruction latches Fault_Active B3:0/4, so the fault remains stored even after the initiating condition disappears.

I used one latched master-fault bit for these machine-level conditions so the rest of the program responds to them in the same way. A temporary E-stop, overload, or mode conflict therefore cannot disappear and allow the conveyor to resume without a deliberate reset.

Ladder Logic

```

+----[ XIO I:1/11 ]-----+
|   Air_Pressure_OK       |
+----[ XIC B3:1/1 ]-----+
|   Jam_Fault             |
|-----+----[ XIC B3:1/2 ]-----+-----( OTL B3:0/4 )----|
|   Reject Timeout Fault   |                               |
|                               |   Fault Active           |

```

Rung 0002 adds the process-related faults to Fault_Active B3:0/4. Loss of Air_Pressure_OK I:1/11, an active Jam Fault B3:1/1, or an active Reject Timeout Fault B3:1/2 makes the rung true and latches the master fault.

I kept these process faults separate from the E-stop and overload rung because they come from different parts of the machine sequence. They still feed the same master fault, so any of them produces the same controlled shutdown while leaving a clear diagnostic path.

Ladder Logic

```
|----[ XIC I:1/2 ]---[ XIC I:1/5 ]---[ XIC I:1/6 ]---[ XIC I:1/11 ]---[ XIO B3:0/3 ]-----+----( OTU B3:0/4 )----|
Reset_PB      EStop_OK      Overload_OK      Air_Pressure_OK      Mode_Conflict      |      Fault_Active
+----( OTU B3:1/1 )----|
|      Jam_Fault
+----( OTU B3:1/2 )----|
Reject Timeout Fault
```

Rung 0003 controls fault recovery. Reset_PB I:1/2 must be ON, EStop_OK I:1/5, Overload_OK I:1/6, and Air_Pressure_OK I:1/11 must all be healthy, and Mode_Conflict B3:0/3 must be OFF. When those conditions are satisfied, the OTU instructions clear Fault_Active B3:0/4, Jam_Fault B3:1/1, and Reject_Timeout_Fault B3:1/2.

I did not allow the operator to clear a fault simply by pressing Reset. The main monitored conditions must first return to a valid state, so Reset confirms recovery instead of temporarily hiding an active problem.

8.5 Rung 0004 — Sequence Recovery Reset

Ladder Logic

```
|----[ XIC I:1/2 ]---[ XIC I:1/5 ]---[ XIC I:1/6 ]--[ XIC I:1/11 ]---[ XIO B3:0/3 ]-----+----( OTU B3:1/0 )----|
      Reset_PB      EStop_OK      Overload_OK      Air_Pressure_OK      Mode_Conflict      |      Reject_Sequence
                                                    +----( OTU B3:0/6 )----|
                                                    |      Product_Tracked
                                                    +----( OTU B3:0/7 )----|
                                                    |      Reject_Product
```

Rung Description:

Rung 0004 uses the same healthy reset conditions as the fault-reset rung, but it clears stored process states. A valid Reset unlatches Reject_Sequence B3:1/0, Product_Tracked B3:0/6, and Reject_Product B3:0/7.

Engineering Purpose:

If the machine is stopped during product handling or a reject cycle, these retained states can remain active even though the physical process has been interrupted. Clearing them during recovery gives the next startup a clean sequence state.

8.6 Rung 0005 — Start Permissive

Ladder Logic

```
|----[ XIO B3:0/4 ]---[ XIC I:1/5 ]---[ XIC I:1/6 ]---[ XIC I:1/11 ]---[ XIO B3:0/3 ]-----+----( OTE B3:0/1 )----|
      Fault_Active      EStop_OK      Overload_OK      Air_Pressure_OK      Mode_Conflict      |      Start_Permissive
```

Rung Description:

Rung 0005 generates Start_Permissive B3:0/1. The bit is true only when Fault_Active B3:0/4 is OFF, EStop_OK I:1/5, Overload_OK I:1/6, and Air_Pressure_OK I:1/11 are ON, and Mode_Conflict B3:0/3 is OFF. Since this is an OTE rung, the permissive changes immediately with the current machine conditions.

Engineering Purpose:

I grouped the main operating conditions into one permissive instead of repeating the same checks throughout the program. The later start and ready logic can use one bit to confirm that the machine is in a valid condition to accept a run command.

8.7 Rung 0006 — Machine Run Request Latch

Ladder Logic

```
|----[ XIC I:1/0 ]---[ XIC B3:0/1 ]-----+----( OTL B3:0/0 )----|
      Start_PB      Start_Permissive      |      Machine_Run_Request
```

Rung Description:

Rung 0006 latches Machine_Run_Request B3:0/0 when Start_PB I:1/0 is pressed while Start_Permissive B3:0/1 is true. The OTL keeps the run request active after the Start pushbutton is released.

Engineering Purpose:

The Start pushbutton is momentary, but the controller needs to remember that the operator requested operation. I store that request internally and keep it separate from the physical conveyor output.

8.8 Rung 0007 — Machine Stop and Fault Stop

Ladder Logic

```
+----[ XIC I:1/1 ]-----+
|   Stop_PB               |
+----[ XIC B3:0/4 ]-----+----( OTU B3:0/0 )----|
|   Fault_Active          |   Machine_Run_Request
+----[ XIO B3:0/1 ]-----+
|   Start_Permissive
```

Rung Description:

Rung 0007 clears Machine_Run_Request B3:0/0 if Stop_PB I:1/1 is pressed, Fault_Active B3:0/4 becomes true, or Start_Permissive B3:0/1 is lost. Any one of the three branches can execute the OTU.

Engineering Purpose:

The stored run request should not survive a stop, a fault, or loss of operating permission. Removing it forces the operator to issue a new Start command after the machine has been returned to a valid condition.

8.9 Rung 0008 — System Ready Status

Ladder Logic

```

                                     +----[ XIC I:1/3 ]----+
                                     |   Auto_Mode         |
|----[ XIC B3:0/1 ]-----[ XIC B3:0/0 ] +-----+-----+
|   Start_Permissive      Machine_Run_Request      |   ( OTE B3:0/2 )----|
|                                     |   System_Ready
|                                     +----[ XIC I:1/4 ]----+
|                                     |   Manual_Mode
```

Rung Description:

Rung 0008 generates System_Ready B3:0/2. Start_Permissive B3:0/1 and Machine_Run_Request B3:0/0 must both be true, and either Auto_Mode I:1/3 or Manual_Mode I:1/4 must be selected. System_Ready drops out as soon as any of those conditions is lost.

Engineering Purpose:

Machine_Run_Request tells me that the operator wants the machine to run; System_Ready confirms that the request is still supported by a valid permissive and a selected operating mode. Keeping those states separate makes the machine status more meaningful.

8.10 Rung 0009 — Automatic Conveyor Request

Ladder Logic

```
|----[ XIC B3:0/2 ]-----[ XIC I:1/3 ]----[ XIO B3:1/0 ]-----+-----+
|   System_Ready      Auto_Mode      Reject_Sequence      |   ( OTE B3:0/5 )----|
|                                     |   Conveyor_Request
```

Rung Description:

Rung 0009 generates the automatic Conveyor_Request B3:0/5. System_Ready B3:0/2 and Auto_Mode I:1/3 must be true, while Reject_Sequence B3:1/0 must be OFF. When a reject sequence begins, the XIO contact opens and the automatic conveyor request drops out.

Engineering Purpose:

The conveyor has to pause during automatic reject handling without shutting down the whole machine. Using Reject_Sequence in this rung stops product movement for the reject operation while System_Ready remains active.

8.11 Rung 0010 — Manual Conveyor Request

Ladder Logic

```
|----[ XIC B3:0/2 ]----[ XIC I:1/4 ]----[ XIC I:1/12 ]----- ( OTE B3:1/3 )----|
      System_Ready      Manual_Mode      Manual_Conveyor_PB      Manual_Conveyor_Request
```

Rung Description:

Rung 0010 controls Manual_Conveyor_Request B3:1/3. The request is true only while System_Ready B3:0/2, Manual_Mode I:1/4, and Manual_Conveyor_PB I:1/12 are all true. Releasing the pushbutton immediately removes the request because the rung uses an OTE.

Engineering Purpose:

I kept manual conveyor movement hold-to-run. Selecting Manual Mode alone cannot start the conveyor; the operator must also hold the dedicated manual conveyor pushbutton.

8.12 Rung 0011 — Conveyor Startup Delay

Ladder Logic

```
      +----[ XIC B3:0/5 ]-----+
      |      Conveyor_Request      |
|----+                               +----[ TON T4:0 ]----|
      |                               |      Delay_Timer
      +----[ XIC B3:1/3 ]-----+      Time Base: 1.0
      |      Manual_Conveyor_Request      |      Preset: 3
```

Rung Description:

Rung 0011 starts Delay_Timer T4:0 from either Conveyor_Request B3:0/5 or Manual_Conveyor_Request B3:1/3. The final ladder uses a 1.0-second time base and a preset of 3, so the timer Done bit turns on after three seconds of continuous request. If both requests disappear before the preset is reached, the TON resets.

Engineering Purpose:

I applied the same startup delay to Automatic and Manual conveyor operation. Both modes therefore reach the motor through the same timing behavior instead of having separate startup logic.

8.13 Rung 0012 — Conveyor Motor Output

Ladder Logic

```
|----[ XIC T4:0/DN ]-----+----[ XIC B3:0/5 ]-----+---- ( OTE O:2/0 )----|
      Delay_Timer_DN      |      Conveyor_Request      |      Conveyor_Motor
                        +----[ XIC B3:1/3 ]-----+
                        |      Manual_Conveyor_Request
```

Rung Description:

Rung 0012 controls Conveyor_Motor O:2/0. T4:0/DN must be true, and either the automatic Conveyor_Request B3:0/5 or Manual_Conveyor_Request B3:1/3 must still be active. Removing the active request turns the motor output off.

Engineering Purpose:

The timer only confirms that the startup delay has elapsed; it does not become a permanent motor command. Rechecking the request at the output keeps the motor tied to the current operating demand.

8.14 Rung 0013 — Product Entry Detection

Ladder Logic

```
|-----[ XIC B3:0/2 ]---[ XIC I:1/3 ]---[ XIO B3:0/6 ]---[ XIC I:1/7 ]----- ( OTL B3:0/6 )-----|  
System_Ready      Auto_Mode      Product_Tracked      Entry_Photoeye      Product_Tracked
```

Rung Description:

Rung 0013 latches Product_Tracked B3:0/6 when the system is ready, Auto Mode is selected, Entry_Photoeye I:1/7 detects a product, and no product is already being tracked. The XIO Product_Tracked contact blocks another entry event until the current product state has been cleared.

Engineering Purpose:

The entry photoeye is only active while a product is physically at that sensor. I latch Product_Tracked so the PLC can remember the product after it leaves the entry station and moves toward the sorting point.

8.15 Rung 0014 — Store Reject Classification

Ladder Logic

```
|--[ XIC B3:0/2 ]---[ XIC I:1/3 ]---[ XIC B3:0/6 ]---[ XIC I:1/7 ]---[ XIC I:1/8 ]----- ( OTL B3:0/7 )--|  
System_Ready      Auto_Mode      Product_Tracked      Entry_Photoeye      Inspection_Reject      Reject_Product
```

Rung Description:

Rung 0014 stores the inspection result for the product currently being tracked. While Product_Tracked B3:0/6 and Entry_Photoeye I:1/7 are true, Inspection_Reject I:1/8 latches Reject_Product B3:0/7. The reject classification remains stored after the product leaves the inspection point.

Engineering Purpose:

The inspection input does not need to remain active all the way to the sorting station. Reject_Product carries the decision with the tracked product until the downstream reject logic uses it.

8.16 Rung 0015 — Product Travel Jam Timer

Ladder Logic

```
|-----[ XIC B3:0/6 ]---[ XIO I:1/9 ]---[ XIC O:2/0 ]-----[ TON T4:1 ]-----|  
Product_Tracked      Sort_Photoeye      Conveyor_Motor      Jam_Timer      Time Base: 1.0      Preset: 120
```

Rung Description:

Rung 0015 runs Jam_Timer T4:1 while Product_Tracked B3:0/6 is true, Sort_Photoeye I:1/9 is still clear, and Conveyor_Motor O:2/0 is running. The final ladder uses a 1.0-second time base and a preset of 120.

Engineering Purpose:

This timer measures how long a tracked product is expected to take to reach the sorting station while the conveyor is actually moving. It gives the PLC a way to recognize a product that has stopped or failed to arrive.

8.17 Rung 0016 — Product Jam Fault

Ladder Logic

```
|--[ XIC T4:1/DN ]---[ XIC B3:0/6 ]---[ XIO I:1/9 ]---[ XIC O:2/0 ]----- ( OTL B3:1/1 )--|  
Jam_Timer_DN      Product_Tracked      Sort_Photoeye      Conveyor_Motor      Jam_Fault
```

Rung Description:

Rung 0016 latches Jam_Fault B3:1/1 when T4:1/DN is true and the original jam conditions are still present: the product remains tracked, Sort_Photoeye is still clear, and the conveyor is running. The OTL keeps the fault stored after the machine shuts down.

Engineering Purpose:

I recheck the process conditions at the point where the fault is generated instead of relying only on the timer Done bit. That makes the jam fault represent a real unresolved travel condition, not just a timer state.

8.18 Rung 0017 — Accepted Product Event

Ladder Logic

```
|----[ XIC B3:0/6 ]-----[ XIC I:1/9 ]-----[ XIO B3:0/7 ]------( OTE B3:1/5 )----|  
Product_Tracked      Sort_Photoeye      Reject_Product      Accepted_Event
```

Rung Description:

Rung 0017 generates Accepted_Event B3:1/5 when a tracked product reaches Sort_Photoeye I:1/9 and Reject_Product B3:0/7 is OFF. Because this is an OTE, Accepted_Event is present only while the accepted-product condition is true.

Engineering Purpose:

I use a separate accepted event so counting and cleanup can be based on a completed sorting result rather than on the entry sensor. It marks the point where a product has actually reached the accepted path.

8.19 Rung 0018 — Start Reject Sequence

Ladder Logic

```
|--[ XIC B3:0/6 ]---[ XIC B3:0/7 ]---[ XIC I:1/9 ]---[ XIO B3:1/0 ]------( OTL B3:1/0 )--|  
Product_Tracked      Reject_Product      Sort_Photoeye      Reject_Sequence      Reject_Sequence
```

Rung Description:

Rung 0018 latches Reject_Sequence B3:1/0 when Product_Tracked B3:0/6 and Reject_Product B3:0/7 are both true and the product reaches Sort_Photoeye I:1/9. The XIO Reject_Sequence contact prevents the rung from repeatedly relatching an already active reject cycle.

Engineering Purpose:

The reject operation has to continue after the sort photoeye condition changes. Latching Reject_Sequence gives the PLC a retained state for the complete reject cycle and also causes the automatic conveyor request to drop out.

8.20 Rung 0019 — Reject Extension Timer

Ladder Logic

```
|----[ XIC B3:1/0 ]-----[ TON T4:2 ]----|  
Reject_Sequence      Reject_Extend_Timer      Time Base: 0.01      Preset: 1000
```

Rung Description:

Rung 0019 runs Reject_Extend_Timer T4:2 whenever Reject_Sequence B3:1/0 is active. The final ladder uses a 0.01-second time base and a preset of 1000, which gives a programmed interval of 10 seconds.

Engineering Purpose:

The reject actuator needs a defined automatic operating interval. T4:2 controls that interval and supplies the Done condition used by the completion and timeout logic.

8.21 Rung 0020 — Manual Reject Request

Ladder Logic

```
|--[ XIC B3:0/2 ]---[ XIC I:1/4 ]---[ XIC I:1/13 ]---[ XIO B3:0/4 ]----- ( OTE B3:1/4 )--|  
    System_Ready      Manual_Mode      Manual_Reject_PB      Fault_Active      Manual_Reject_Request
```

Rung Description:

Rung 0020 generates Manual_Reject_Request B3:1/4 while System_Ready B3:0/2, Manual_Mode I:1/4, and Manual_Reject_PB I:1/13 are true and Fault_Active B3:0/4 is OFF. Releasing the pushbutton removes the request immediately.

Engineering Purpose:

Manual reject operation is hold-to-run and remains blocked during a master fault. This lets the actuator be tested in Manual Mode without allowing the mode selector alone to energize it.

8.22 Rung 0021 — Reject Solenoid Output

Ladder Logic

```
      +-----[ XIC B3:1/0 ]-----[ XIO T4:2/DN ]-----+  
      |      Reject_Sequence      Reject_Timer_DN      |  
|-----+                                         +-----[ XIO B3:0/4 ]----- ( OTE O:2/1 )-----|  
      |                                         |      Fault_Active      Reject_Solenoid  
      +-----[ XIC B3:1/4 ]-----+  
      Manual_Reject_Request
```

Rung Description:

Rung 0021 controls Reject_Solenoid O:2/1 from two possible command paths. In Automatic operation, Reject_Sequence B3:1/0 keeps the solenoid energized while T4:2/DN is false. Manual_Reject_Request B3:1/4 provides the alternate manual path. Fault_Active B3:0/4 must remain OFF for either path to energize the output.

Engineering Purpose:

I kept both Automatic and Manual reject commands in one physical output rung. That avoids duplicate coils and makes the master fault interlock apply to the actuator regardless of which mode is requesting it.

8.23 Rung 0022 — Reject Complete Event

Ladder Logic

```
|--[ XIC B3:1/0 ]-----[ XIC T4:2/DN ]-----[ XIC I:1/10 ]----- ( OTE B3:1/6 )--|  
    Reject_Sequence      Reject_Timer_DN      Reject_Confirm      Reject_Complete_Event
```

Rung Description:

Rung 0022 generates Reject_Complete_Event B3:1/6 when Reject_Sequence B3:1/0 is active, T4:2/DN is true, and Reject_Confirm I:1/10 is present. The event indicates that the timed reject cycle completed and the expected confirmation was received.

Engineering Purpose:

I do not count a reject only because the solenoid was commanded. The confirmation input has to prove that the reject operation completed before the sequence is treated as successful.

8.24 Rung 0023 — Reject Confirmation Timeout Fault

Ladder Logic

```
|--[ XIC B3:1/0 ]----[ XIC T4:2/DN ]----[ XIO I:1/10 ]----- ( OTL B3:1/2 )--|
    Reject_Sequence    Reject_Timer_DN    Reject_Confirm          Reject_Timeout_Fault
```

Rung Description:

Rung 0023 latches Reject_Timeout_Fault B3:1/2 when the reject sequence is active and T4:2 is done but Reject_Confirm I:1/10 is still absent. The XIO confirmation contact distinguishes this condition from a successful reject completion.

Engineering Purpose:

This fault identifies a reject cycle that was commanded but not confirmed. It prevents an unsuccessful reject from being treated like normal production and sends the condition into the master fault path.

8.25 Rung 0024 — Total Product Counter

Ladder Logic

```
      +----[ XIC B3:1/5 ]-----+
      |      Accepted_Event      |
|----+                               +----[ CTU C5:0 ]----|
      |                               |      Total_Product_Count
      +----[ XIC B3:1/6 ]-----+      Preset: 9999
      |      Reject_Complete_Event
```

Rung Description:

Rung 0024 increments Total_Product_Count C5:0 from either Accepted_Event B3:1/5 or Reject_Complete_Event B3:1/6. The two event paths represent the valid completed outcomes for a processed product.

Engineering Purpose:

I count products after they reach a completed sorting result rather than when they first enter the machine. This makes the total represent processed products, not just detections at the entry sensor.

8.26 Rung 0025 — Accepted Product Count

Ladder Logic

```
|----[ XIC B3:1/5 ]-----[ ADD ]----|
    Accepted_Event          Source A: N7:0
                              Source B: 1
                              Dest:    N7:0
```

Rung Description:

Rung 0025 uses an ADD instruction to increase N7:0 by one when Accepted_Event B3:1/5 is true. The result is written back to N7:0, maintaining the accepted-product total.

Engineering Purpose:

I kept an accepted count separate from the overall product counter so the program can report how many completed products passed inspection.

8.27 Rung 0026 — Rejected Product Count

Ladder Logic

```
|----[ XIC B3:1/6 ]-----[ ADD ]----|
      Reject_Complete_Event      Source A: N7:1
                                   Source B: 1
                                   Dest:   N7:1
```

Rung Description:

Rung 0026 uses an ADD instruction to increase N7:1 by one when Reject_Complete_Event B3:1/6 is true. Because the event requires Reject_Confirm, only successfully confirmed rejects are added to this total.

Engineering Purpose:

The rejected-product total should represent completed reject actions, not failed attempts. Tying the count to Reject_Complete_Event keeps the production statistics consistent with the reject verification logic.

8.28 Rung 0027 — Accepted Product Cleanup

Ladder Logic

```
|----[ XIC B3:1/5 ]-----+----- ( OTU B3:0/6 )----|
      Accepted_Event      |           Product_Tracked
                        +----- ( OTU B3:0/7 )----|
                        |           Reject_Product
```

Rung Description:

Rung 0027 clears Product_Tracked B3:0/6 and Reject_Product B3:0/7 when Accepted_Event B3:1/5 is true. These OTU instructions execute after the counting rungs in the program.

Engineering Purpose:

I placed accepted-product cleanup after counting so the event can update the production totals before the stored product state is removed. The cleanup then prepares the tracking logic for the next incoming product.

8.29 Rung 0028 — Rejected Product Cleanup

Ladder Logic

```
|----[ XIC B3:1/6 ]-----+----- ( OTU B3:1/0 )----|
      Reject_Complete_Event |           Reject_Sequence
                        +----- ( OTU B3:0/6 )----|
                        |           Product_Tracked
                        +----- ( OTU B3:0/7 )----|
                        |           Reject_Product
```

Rung Description:

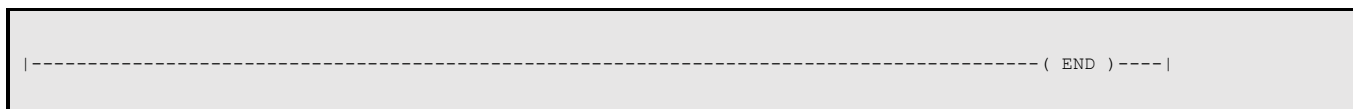
Rung 0028 clears Reject_Sequence B3:1/0, Product_Tracked B3:0/6, and Reject_Product B3:0/7 when Reject_Complete_Event B3:1/6 is true. The OTU instructions remove the retained states associated with the completed reject cycle.

Engineering Purpose:

A confirmed reject should finish both the actuator sequence and the product-tracking sequence. Clearing these bits together returns the sorting logic to its idle state and allows the next product to be tracked normally.

8.30 Rung 0029 — END

Ladder Logic



Rung Description:

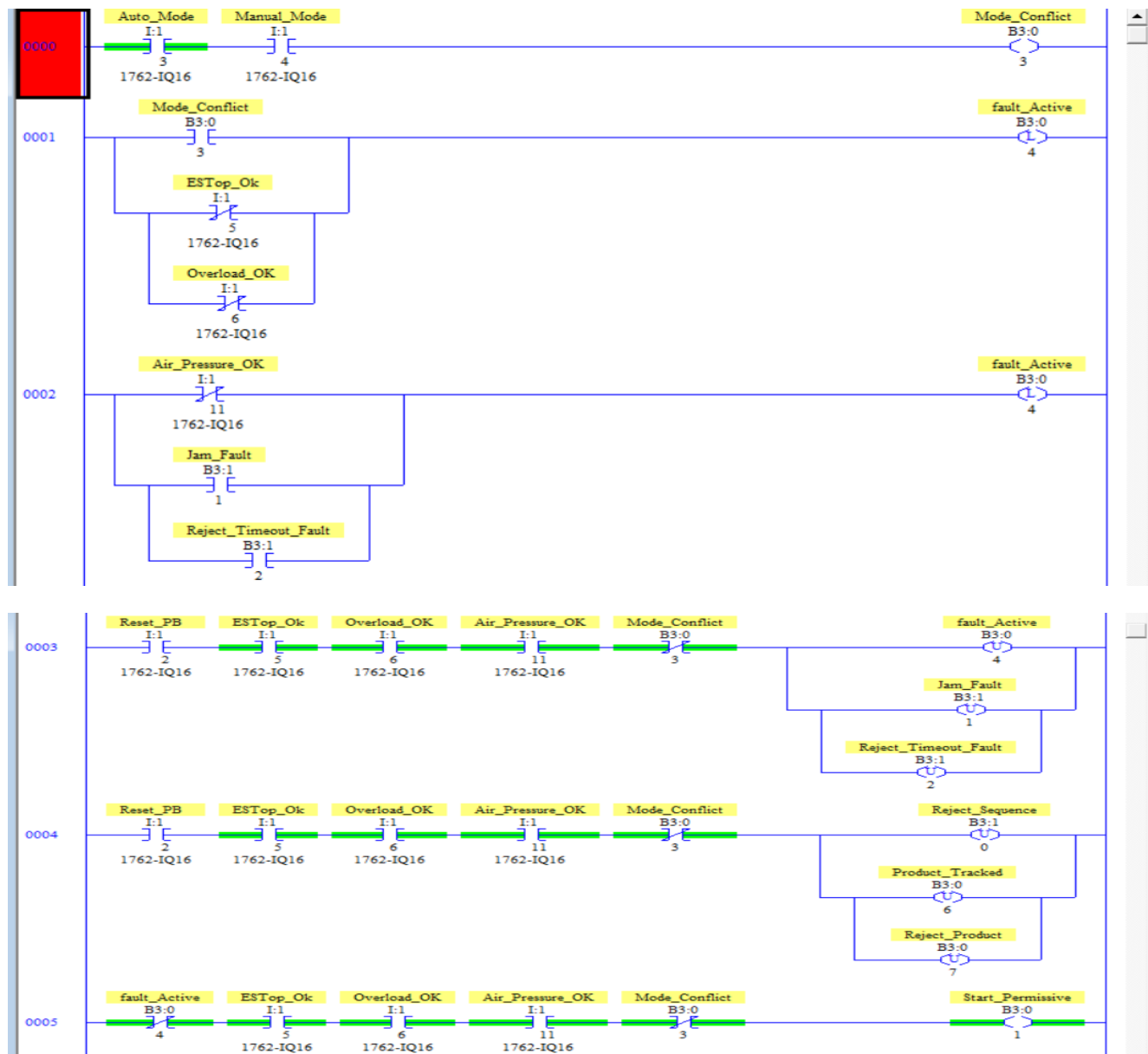
Rung 0029 contains the END instruction, which marks the end of the main ladder file.

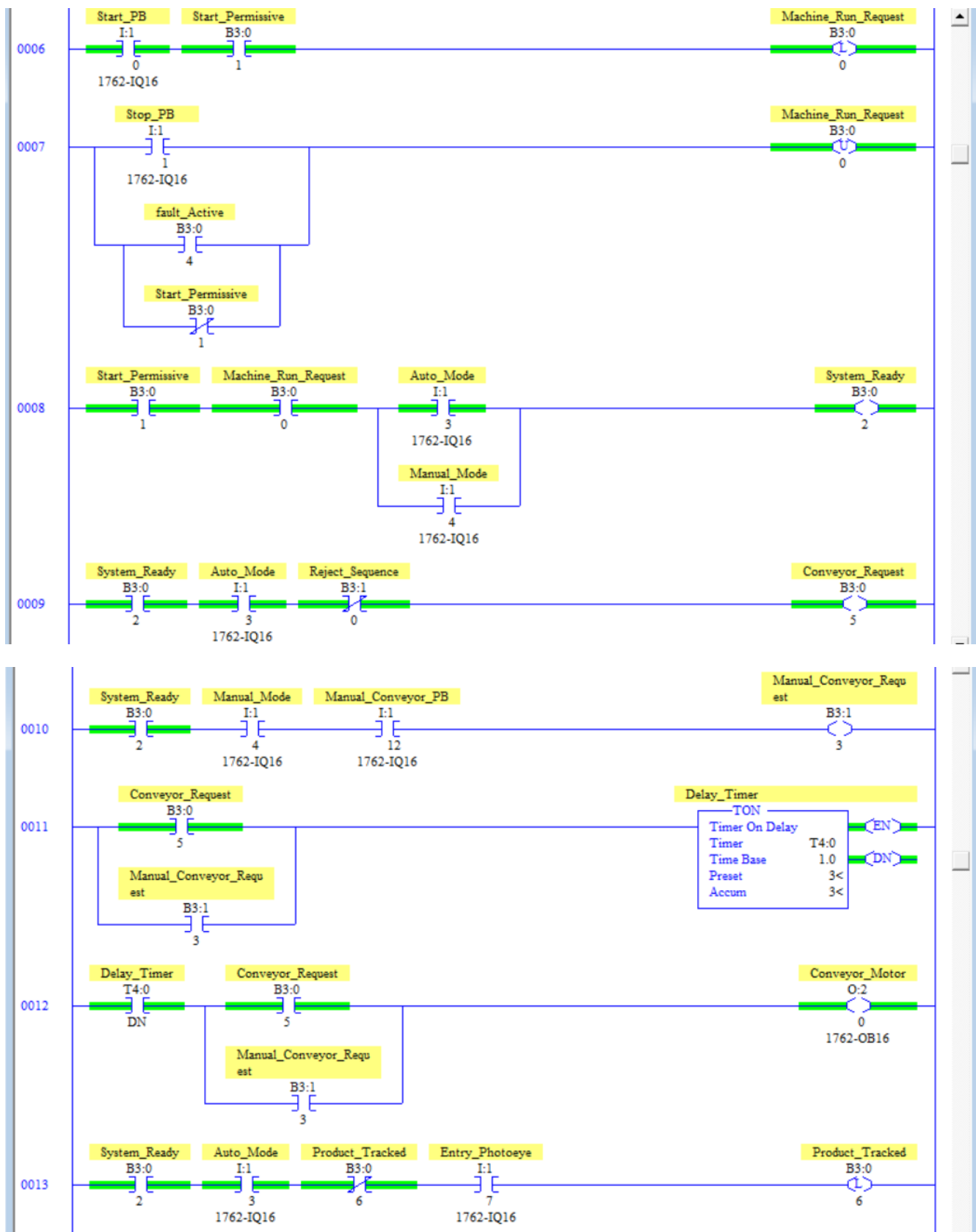
Engineering Purpose:

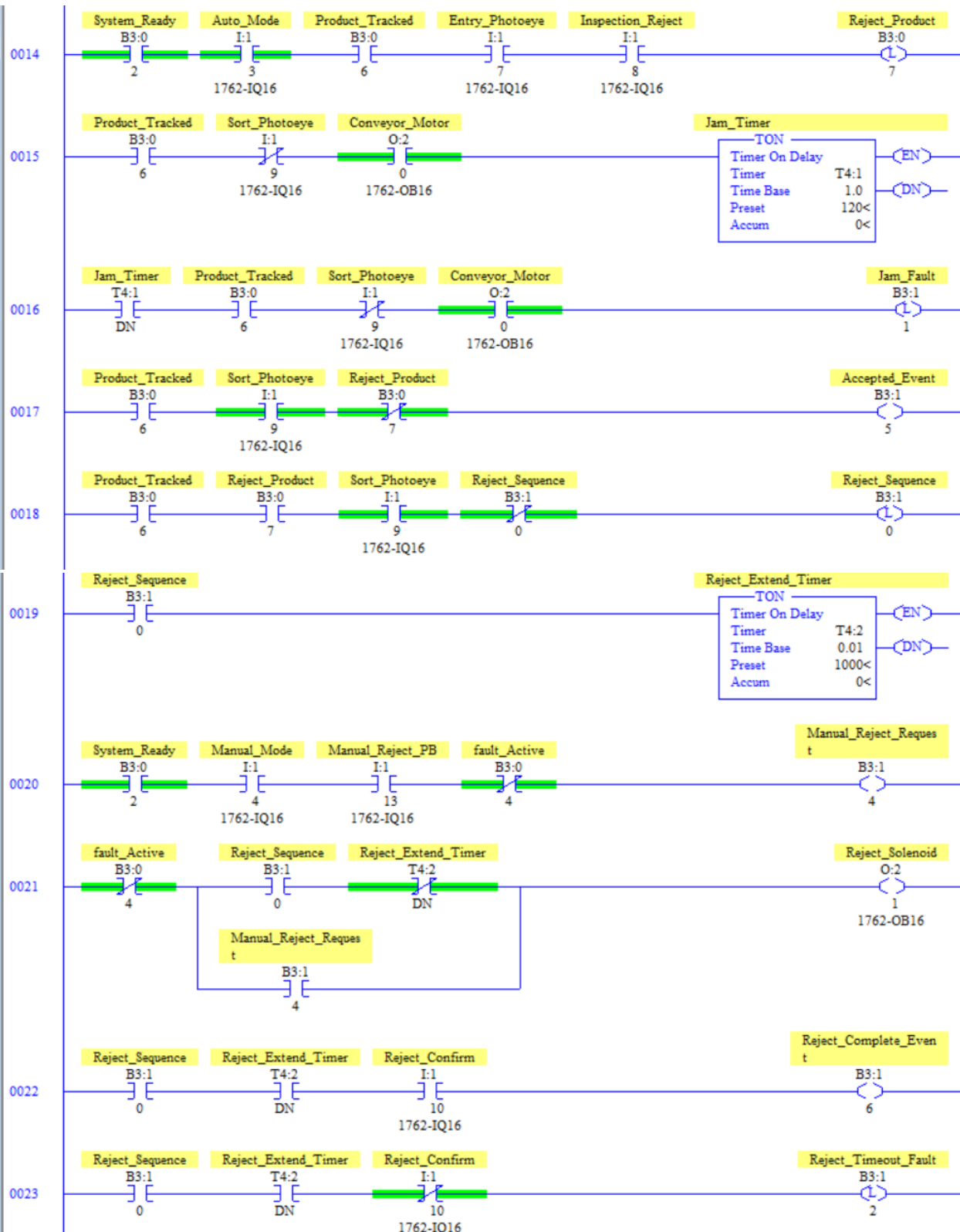
This is the program boundary for the final Project 6 ladder. No control logic in this file executes below the END instruction.

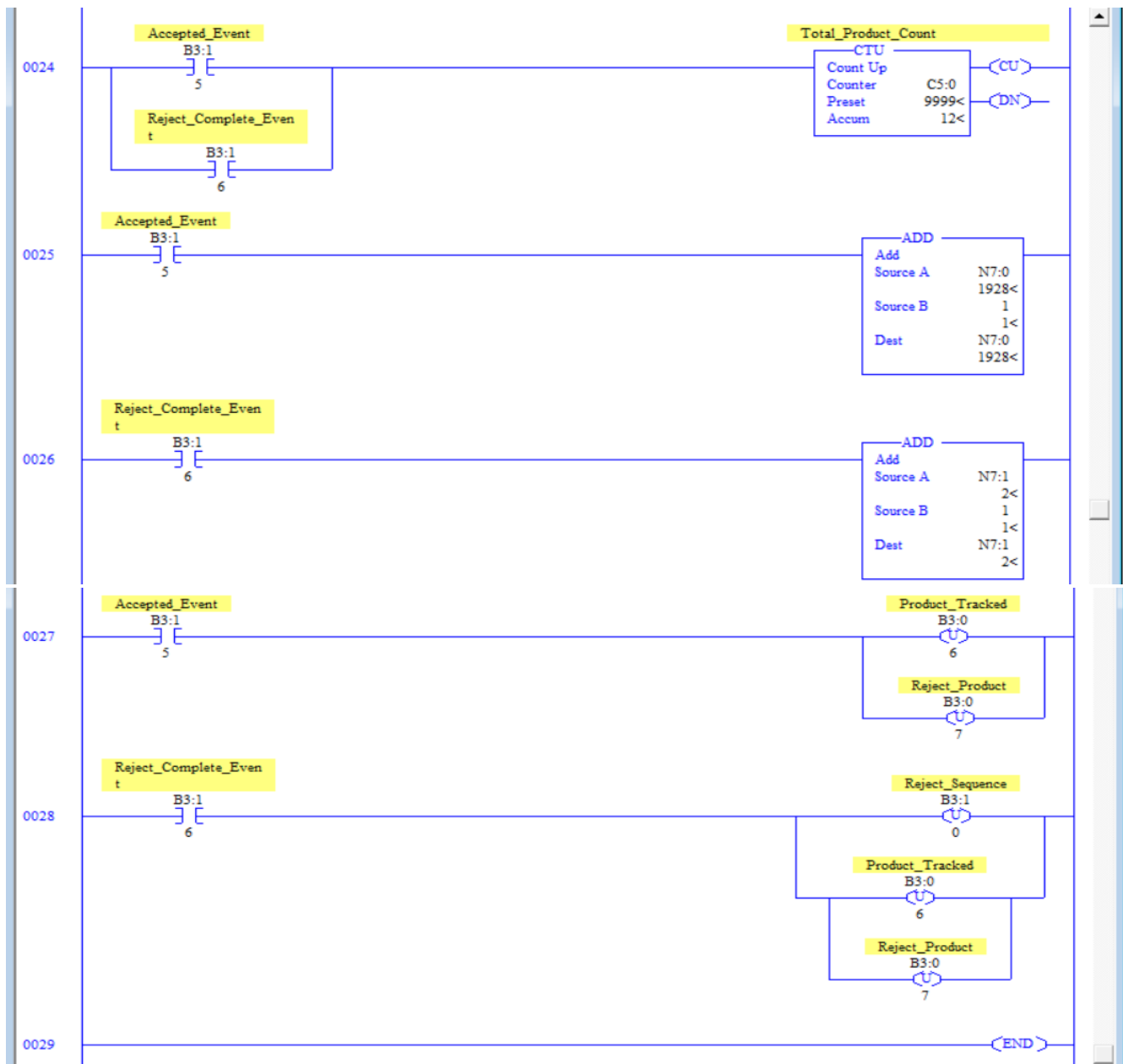
8.31 Actual RSLogix Implementation

The following images are the final Project 6 Version 2 ladder screenshots and are the technical source of truth for this portfolio. The typed rung representations above include the screenshot-confirmed PLC addresses and symbolic names. If any visual ambiguity remains, the final RSLogix screenshot governs.









9. Sequence of Operation

- Step 1 — The PLC evaluates mode selection, E-stop status, overload status, air pressure, and stored process faults.
- Step 2 — Healthy conditions establish Start_Permissive B3:0/1.
- Step 3 — Start_PB I:1/0 latches Machine_Run_Request B3:0/0.
- Step 4 — With a valid Auto or Manual mode selected, System_Ready B3:0/2 becomes true.
- Step 5 — In Auto mode, Conveyor_Request B3:0/5 starts the automatic conveyor sequence when Reject_Sequence is inactive.
- Step 6 — T4:0 provides the programmed three-second startup delay before Conveyor_Motor O:2/0 energizes.
- Step 7 — Entry_Photoeye I:1/7 latches Product_Tracked B3:0/6 for the incoming product.
- Step 8 — Inspection_Reject I:1/8 can latch Reject_Product B3:0/7 for that tracked product.
- Step 9 — Jam_Timer T4:1 monitors travel time to Sort_Photoeye I:1/9.

Step 10 — A tracked accepted product at Sort Photoeye generates Accepted_Event B3:1/5.

Step 11 — A tracked reject product at Sort Photoeye latches Reject_Sequence B3:1/0 and removes automatic conveyor demand.

Step 12 — T4:2 controls the screenshot-confirmed 10-second reject interval.

Step 13 — Reject_Solenoid O:2/1 is controlled by the automatic reject sequence or Manual_Reject_Request B3:1/4 while Fault_Active is false.

Step 14 — Reject_Confirm I:1/10 after timer completion generates Reject_Complete_Event B3:1/6.

Step 15 — Missing Reject Confirm at timer completion latches Reject_Timeout_Fault B3:1/2.

Step 16 — Accepted_Event or Reject_Complete_Event updates Total_Product_Count C5:0.

Step 17 — Accepted_Event increments N7:0 and Reject_Complete_Event increments N7:1.

Step 18 — Cleanup rungs clear the stored product/reject states after counting.

10. Testing & Validation

Test	Verified Function	Result
Startup	Ready state and delayed conveyor start	PASS
Stop	Run request removed and conveyor stopped	PASS
Safety / Operating Faults	E-stop, overload, air pressure and mode conflict	PASS
Fault Recovery	Controlled reset after healthy conditions restored	PASS
Good Product	Tracking, accepted event and counting	PASS
Reject Product	Classification, reject sequence, confirmation and counting	PASS
Jam Detection	Jam Fault generated for excessive travel time	PASS
Reject Timeout	Fault generated when confirmation was missing	PASS
Manual Mode	Manual conveyor and manual reject functions	PASS

The supplied testing report records successful RSLogix online validation of the final PLC program. These results are documented as PLC/simulation validation, not physical conveyor commissioning or machine-safety certification.

11. Results

Project 6 Version 2 successfully combines machine operation, product tracking, classification, reject sequencing, confirmation monitoring, production counting, and fault recovery in one structured PLC application.

A major result is that product information persists between physical sensing locations. Product_Tracked retains product presence, while Reject_Product retains its classification. The program also distinguishes actuator command from process verification: Reject_Solenoid represents what the PLC requested, while Reject_Confirm provides evidence that the expected reject result occurred.

Jam_Fault and Reject_Timeout_Fault add process diagnostics beyond the basic machine-health inputs, while C5:0, N7:0, and N7:1 provide total, accepted, and rejected production information.

12. Challenges & Troubleshooting

Symptom	Diagnostic Path
System will not become ready	B3:0/4 → I:1/5 → I:1/6 → I:1/11 → B3:0/3 → B3:0/1 → B3:0/0 → mode inputs
Conveyor does not start	B3:0/5 or B3:1/3 → T4:0/DN → O:2/0
Product not tracked	B3:0/2 → I:1/3 → I:1/7 → B3:0/6
Reject classification not stored	B3:0/6 → I:1/7 → I:1/8 → B3:0/7
Jam fault occurs	B3:0/6 → I:1/9 → O:2/0 → T4:1 → B3:1/1
Reject sequence does not start	B3:0/6 → B3:0/7 → I:1/9 → B3:1/0
Reject solenoid does not energize	B3:1/0 / B3:1/4 → T4:2/DN → B3:0/4 → O:2/1
Reject timeout occurs	B3:1/0 → T4:2/DN → I:1/10 → B3:1/2
Accepted count incorrect	B3:1/5 → C5:0 / N7:0 → cleanup order
Rejected count incorrect	B3:1/6 → C5:0 / N7:1 → cleanup order

The main troubleshooting lesson is to trace the current machine state from permissives through the active process step before inspecting the physical output. Product-tracking systems depend on stored states, so a sensor can be physically clear while the PLC correctly retains product or reject information.

13. Lessons Learned

This project strengthened my understanding of state-based product tracking. A field sensor only represents the current physical condition; a latched internal state lets the PLC remember product information after the product leaves that sensor.

I also learned the importance of retaining the inspection result separately from the inspection input. Reject_Product carries the decision to the sorting station. Reject confirmation further reinforced the difference between commanding an actuator and verifying that the process actually completed.

The counting and cleanup arrangement also demonstrated deliberate use of PLC scan order: production counts are updated before the event-enabling states are cleared.

14. Industrial Relevance

Conveyor inspection and sorting systems are widely used in packaging, manufacturing, logistics, food processing, and automated material handling. This project demonstrates relevant controls concepts including product tracking between stations, retained classification, Auto/Manual operation, startup permissives, process timers, reject sequencing, actuator confirmation, jam detection, production counting, and controlled fault recovery.

A physical industrial implementation could additionally require VFD control, motor-running feedback, safety-rated devices, pneumatic pressure instrumentation, multiple-product tracking, encoder-based positioning, barcode or vision inspection, HMI alarm history, and production data logging.

15. Version 1 → Version 2 Engineering Progression

Version 1 Foundation	Version 2 Implementation	Engineering Improvement
Basic conveyor operation	Machine Run Request + System Ready	Separates command from authorization
Simple start/stop	Start Permissive + latched faults	Structured operating control
Single operating behavior	Auto/Manual architecture	Production and maintenance modes
Direct motor control	Conveyor requests + startup timer	Separates sequence command from output
Simple sensor reaction	Product_Tracked state	Remembers product between stations
No retained classification	Reject_Product latch	Carries inspection result downstream

Basic conveyor sequence	Sorting-station processing	Adds product-dependent behavior
No travel monitoring	Jam Timer + Jam Fault	Detects abnormal product movement
No reject verification	Reject Confirm	Verifies process completion
No reject-failure diagnostic	Reject Timeout Fault	Detects missing confirmation
Basic counting	Total + accepted + rejected counts	Provides production statistics
Limited recovery	Fault and sequence cleanup	Supports controlled restart
Direct reject actuation	Reject sequence + timer + confirmation	Creates defined reject cycle

16. Project Summary

Project 6 Version 2 — Industrial Conveyor Sorting System extends the original conveyor concept into a structured automatic sorting application. The final ladder contains Rungs 0000–0028 plus END at 0029 and implements mode management, machine-health monitoring, Start Permissive, Machine Run Request, System Ready, automatic and manual conveyor control, startup delay, product tracking, stored reject classification, jam monitoring, accepted-product handling, reject sequencing, manual reject operation, reject confirmation, reject timeout detection, and production counting.

Product_Tracked B3:0/6 retains product presence and Reject_Product B3:0/7 retains classification while the product travels between stations. Accepted products generate Accepted_Event B3:1/5, while successfully confirmed rejects generate Reject_Complete_Event B3:1/6. Jam_Fault B3:1/1 and Reject_Timeout_Fault B3:1/2 provide process diagnostics beyond the basic machine-health inputs.